

31. BICARBONATE BEFORE HIT TRAINING BOOSTS TRAINING ADAPTATIONS

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When exercising at high intensity (above second threshold/FTP) a lot of **hydrogen ions (H^+)** are produced in the working muscles. Without going too deep into the physiological mechanisms, H^+ accumulation leads to muscle fatigue and a decrease in performance. **Bicarbonate (HCO_3^-) is a molecule acting as H^+ buffer (reducing H^+ concentration) both in the muscle and in the blood.** For this reason, **bicarbonate supplementation is widely used as an acute performance enhancer to ingest before/during competitions** with efforts above the second threshold/FTP/critical power.

Interestingly, we have a study suggesting bicarbonate can also be used chronically (for weeks) before high intensity training sessions to boost physiological training adaptations and so, ultimately, performance in the race day. (1)

..How?

High Intensity Training (HIT) represents performing a workout with intervals above the second threshold/FTP/critical power, so it generally leads to a net H⁺ accumulation. HIT is known to be a strong stimulus to induce aerobic physiological adaptations which benefits performance. In particular, it stimulates mitochondrial (the muscles' powerhouses where energy is produced through the aerobic system) biogenesis and growth. This process is activated and powered by PGC-1 α , a transcriptional coactivator factor. **High intensity training triggers a lot of changes in the cellular muscular environment which globally activate PGC-1 α and so the positive aerobic physio and performance adaptations. However, one of the environmental changes provoked, if taken alone, would inhibit PGC-1 α : this is H⁺ accumulation!** In fact, a previous study (2) showed that ingesting a H⁺ supplement before HIT sessions blunted the post-exercise response of PGC-1 α !

So, while HIT globally induce positive mitochondrial adaptations, we could maximise its effects reducing the H⁺ net accumulation that is naturally associated with this training intensity. Here enter supplementing with bicarbonate (H⁺buffer) chronically (for weeks) before HIT.

Does chronic (for weeks) bicarbonate supplementation before high intensity training sessions boost training adaptations and so performance?

A study published by Edge and colleagues in 2006 on *Journal of Applied Physiology* can help to answer this question. (1)

HOW DID THEY TRAIN?

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- 16 recreationally active but not competitive cyclists female students were recruited.
 - They were divided into two groups (BICARB s CONTROL) which for 8 weeks performed the same training schedule: 3 days/week of 6-12 x 2min at 140 –170% of second threshold.
 - The only differences between the two groups was:
 - **BICARB: ingested 0.4 g/kg of sodium bicarbonate ~1h before each HIT session (0.2 g/kg 90min before + 0.2 g/kg 30min before). For example, this dose represents 28g for a 70kg athlete.**
 - CONTROL: ingested a placebo drink at the same timepoints

WHAT DID THEY FIND?

Physiological and performance changes between pre-post the 8 training weeks were globally superior when assuming BICARB vs CONTROL:

- VO2max: +22% vs +17%
- **Power at second threshold: +26% vs +15% (statistically significant!)**
- **Time to task failure at 85% of VO2max measured before the 8-weeks** (the effort lasted around 10-15 min): **164% vs +123% (statistically significant!)**

PRACTICAL TAKEAWAYS

Ingesting sodium bicarbonate (0.4 g/kg) before HIT sessions for some weeks can boost training adaptations and performance.

GABRIELE'S COMMENT

- Aside the widely used acute bicarbonate supplementation to enhance performance during race day, this study suggests that bicarbonate can also be ingested chronically (for weeks) before high-intensity training sessions to boost the physiological aerobic adaptations derived from training stimulus.
- A possible limitation of this study is that participants were just 'recreationally active'. So, we are not sure that the same results can be obtained with world class athletes. However, I can't see a physiological reason why this should not work with higher level athletes: they also accumulated H⁺ above second threshold/FTP/critical power.
- Note that bicarbonate supplementation could create some gastrointestinal issue, so it is important to choose the right product on the market and make some trials. We will come back on this in the next posts!

Thanks for Reading! Spread with your Team!

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